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Shravan Chaudhari

Johns Hopkins University | NYU Courant | BITS Pilani

Personal Website
Shravan-Chaudhari
Google Scholar

Education

Doctor of Philosophy (Ph.D.), Computer Science, Johns Hopkins University (Whiting School of Engineering) **May 2023 - Present**
Advisors: Dr. Rama Chellappa & Dr. Suchi Saria
Cumulative GPA: 4.0/4.0

Master of Science, Computer Science, New York University (Courant Institute of Mathematical Sciences) **Sep 2021 — May 2023**
Cumulative CGPA: 3.83/4.0

Bachelor of Engineering, Electronics & Instrumentation, Birla Institute of Technology & Science Pilani **Aug 2017 — May 2021**
Cumulative GPA: 8.52/10.0

Research Interests

Trustworthy and Fairness in AI, LLM Agents, Graph informed RAGs, Domain Adaptation, Out-Of-Distribution Detection.

Industry Experience

Student Researcher, Microsoft (CoreAI GitHub Copilot Science team) **Jan 2026 — Present**
- Improved LLM based code repository navigation and retrieval for AI coding agents: Existing scaffoldings exhaust the context of small cost-efficient LLMs, while restrictive ones lose the structural awareness needed for repository navigation.
- Built two complementary solutions: (a) GRPO RL fine-tuning of Qwen2.5-Coder 7B/14B on the RepoSearcher scaffolding with DICE rewards on SWE-Bench Lite, and (b) a graph-aware scaffolding constraining agents to three bounded actions over a code graph—dense top-K, BM25 top-K, and neighborhood BFS.
- **Results:** GRPO improved file-level Recall@1 by $\sim$13% on the 14B (matching a 30B training-free baseline) and by $\sim$40% on the 7B; the graph-aware scaffolding matches this at small K without any RL training and exhaustion of the LLM's context.

Applied Scientist II Intern, Amazon Inc. (Coding Agents Team) **May 2025 — Sep 2025**
- Developed and implemented an LLM-based dense embedding retrieval algorithm to localize relevant code snippets from large production codebases given GitHub-style issue descriptions.
- Built graph representations of large codebases and an LLM-driven agent framework for code navigation, leveraging dense embeddings over traditional BM25 sparse retrieval methods.
- **Results:** Improved existing retrieval performance (Recall@K) by 20%+ consistently across Python, Java, JavaScript, and TypeScript, leading to higher pass rates in solving GitHub-style issues. Publication under review .

Applied Scientist II Intern, Amazon Inc (Search Experience Team) **Aug 2024 — Dec 2024**
- Designed a multimodal LLM-based feedback mechanism and novel metric to measure the impact of Amazon search pages on customer experience and engagement.
- Applied foundation models (Claude Sonnet 3.0, Llama 3.2, SAM2) to evaluate product images, descriptions, and metadata for search page quality assessment.
- **Key results:** Achieved 72% correlation with human annotators, validating the LLM feedback mechanism. Built an end-to-end multimodal framework that recommends search page improvements informed by principles of psychology and cognitive science. Publication

Machine Learning & Software Co-Op, Johnson & Johnson MedTech (Algorithms & AI team) **May 2022 — Dec 2022**
- Developed efficient data-centric semi-supervised active learning framework for minimally invasive surgical robots to automatically curate only the most useful task-specific video clips from hundreds of 3-4 hours long medical surgery videos.
- Developed novel data selection algorithms to identify anomalies and ambiguous data to mitigate computer vision model uncertainties & improve the performance of biomedical tool detection & instance segmentation models used in robots during surgical processes.
- **Key results:** Improved the performance of biomedical tool detection and instance segmentation with 70%+ reduction in annotation costs.

Publications

1. **Shravan S Chaudhari**, Jacob, R. T., Rashid, S., Goswami, M. & Bock, C. *SpIDER: Spatially Informed Dense Embedding Retrieval of Software Bugs in the Web of Code Graphs (In preparation to be submitted at ICML 2026)*
2. **Shravan Chaudhari**, Wald, Y. & Saria, S. *Open-Set Domain Adaptation Under Background Distribution Shift: Challenges and A Provably Efficient Solution* 2025. arXiv: 2512.01152 `[cs.LG]`. https://arxiv.org/abs/2512.01152.
3. **Shravan Chaudhari**, Akula, T., Kim, Y. & Blake, T. *Multimodal LLM Augmented Reasoning for Interpretable Visual Perception Analysis* in *CHI 2025 Workshop on Human-AI Interaction for Augmented Reasoning* (2025). https://arxiv.org/abs/2504.12511.
4. Kaushik, P., **Shravan Chaudhari**, Vaidya, A., Chellappa, R. & Yuille, A. *The Universal Weight Subspace Hypothesis* 2025. arXiv: 2512.05117 `[cs.LG]`. https://arxiv.org/abs/2512.05117.
5. Kaushik, P., Vaidya, A., **Chaudhari, Shravan** & Yuille, A. *EigenLoRAx: Recycling Adapters to Find Principal Subspaces for Resource-Efficient Adaptation and Inference* in *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR) Workshops* (June 2025), 649–659.

6. Chung, H.-H., **Shravan S Chaudhari**, Han, X., Wald, Y., Saria, S. & Ghosh, J. Between Linear and Sinusoidal: Rethinking the Time Encoder in Dynamic Graph Learning. *Transactions on Machine Learning Research.* ISSN: 2835-8856. https://openreview.net/forum?id=W6GQvdOGHg (2025).
7. Chung, H.-H., **Chaudhari, Shravan**, Wald, Y., Han, X. & Ghosh, J. *Novel Node Category Detection Under Subpopulation Shift* in *Machine Learning and Knowledge Discovery in Databases. Research Track* (eds Bifet, A., Davis, J., Krilavičius, T., Kull, M., Ntoutsi, E. & Žliobait\.e, I.) (Springer Nature Switzerland, Cham, 2024), 196–212. ISBN: 978-3-031-70359-1.
8. Singh, P., Chukkapalli, R., **Chaudhari, Shravan**, Chen, L., Chen, M., Pan, J., Smuda, C. & Cirrone, J. Shifting to machine supervision: annotation-efficient semi and self-supervised learning for automatic medical image segmentation and classification. *Scientific Reports* **14,** 10820. ISSN: 2045-2322. https://doi.org/10.1038/s41598-024-61822-9 (May 2024).
9. Andrews, M., Burkle, B., Croce, D. D., **Shravan Chaudhari**, Gleyzer, S., Heintz, U., Naraina, M., Paulini, M. & Usai, E. Accelerating End to End Deep Learning for Particle Reconstruction using CMS open data. *The European Physical Journal Conferences 251:03057* (2021).
10. Andrews, M., Burkle, B., Croce D. D., **Shravan Chaudhari**, Gleyzer, S., Heintz, U., Naraina, M., Paulini, M. & Usai, E. End-to-End Jet Classification of Boosted Top Quarks with CMS Open Data. *The European Physical Journal Conferences 251:04030* (2021).
11. Singh, P., Chen, L., Chen, M., Pan, J., Chukkapalli, R., **Chaudhari, Shravan** & Cirrone, J. *Enhancing Medical Image Segmentation: Optimizing Cross-Entropy Weights and Post-Processing with Autoencoders* in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops* (Oct. 2023), 2684–2693.
12. Chaudhari, Purva, **Chaudhari, Shravan**, Chudasama, Ruchi & Gleyzer, Sergei. End-to-end deep learning inference with CMSSW via ONNX using Docker. *EPJ Web of Conf.* **295,** 09015. https://doi.org/10.1051/epjconf/202429509015 (2024).

Research Experience

Research Assistant @ Center for Data Science, New York University (Advisor: Dr. Jacopo Cirrone) — **Jan 2023 — May 2023**

- Developed novel zero shot and few shot segmentation and classification algorithms using semi and self-supervised representation learning for histopathology images from unknown distributions.
- Links to publications (Nature Scientific Reports) (ICCV 2023)

Research Fellow @ National Science Foundation, Princeton Institute of Computational Science and Engineering — **May 2021 — Oct 2021**

- Developed graph based representations of high energy particle (Taus) to identify particle collision energy signals from 3 different types of background noises.
- Worked with graph neural networks to perform the classification between the signals and background noises using SAGEConv functions on generated graph representations.
- Performed a detailed analysis and comparison of graph neural networks, vision transformers and sparse CNNs.
- **Key accomplishments:** Improved the ROC-AUC score on benchmark dataset from 0.7 to 0.85 using GNNs for Tau particle identification which outperformed vision transformers and state of the art ML algorithms. .

Student Researcher @ CERN, Undergraduate Thesis — **Sep 2020 — May 2021**

- Developed novel deep convolutional neural network (DCNNs) based solutions for classification and reconstruction of high energy particles like electrons, photons, quarks, gluons and top jets.
- Performed detailed study analysing the strengths and weaknesses of ResNet based DCNNs on different type of particle image channels like electromagnetic calorimeter, hadronic calorimeter and characteristics of particles such as location of the energy deposit and transverse momentum recorded by silicon detectors at the Large Hadron Collider.
- **Key accomplishments:** Beat the state-of-the-art performance by achieving at least 95% accuracy for each of the particle categories based on raw deposit images. It lead to couple of publications that made this research a pioneer in establishing the use of using deep learning technique for particle identification and reconstruction during the run of the experiments at CERN.

Mentoring and Advising Experience

- **Mentor and Organizer @ Machine Learning for Science organisation:** Mentored 3 students through google summer of code 2022 program on machine learning projects for astrophysics & particle physics .

Open Source Developer Experience

Google Summer of Code 2021, — **May 2021 — Aug 2021**

- Designed & implemented graph based modelling strategies to represent and classify low momentum particle images having 13 energy channels recorded by various silicon detectors at the Large Hadron Collider.
- Developed a standard framework to deploy graph neural networks (GNNs) within CERN software pipeline using C++ ONNX API and Torch-Geometric library.
- **Key accomplishments:** Benchmarked the graph neural network performance and its deployment using the newly built framework for classification of electrons, photons, quark, gluons and boosted top jets by achieving ROC AUC score of atleast 0.95 for each category

Google Summer of Code 2020, **May 2020 — Aug 2020**

- Created a C++ based End-to-End Software (E2E) framework to enable advanced data processing and complex analysis on the CERN database.
- **Key Accomplishments:** Integrated the E2E framework with the CERN inference engine to support deployment of machine learning architectures like GNNs, CNNs, Variational Autoencoders, etc. trained with either of the 4 different frameworks: tensorflow, mxnet, onnx and pytorch.

International Nvidia Hackatons 2020-2021, Nvidia CSCS Hackathon, GPU Helmholtz hackathon

- Optimised and parallelized the training and inference of large scale ResNets and graph neural networks on huge CERN dataset containing 4 million particle images each having 8 channels using computing nodes and V100 GPUs provided by Nvidia as part of their International GPU Hackathons, Bootcamps and other collaborations .
- Used horovod framework for optimising tensorflow based ResNet models and tensorRT SDK for optimising the pytorch model inferences.
- **Key Accomplishments:** Accelerated the ResNet training speed by 93% on optimising for a single V100 GPU and by 97% by distributing the model training across 4 V100 GPUs.

Talks

- Talk on Accelerating end-to-end deep learning for high energy particle identification using Graph Neural Networks, National Science Foundation IRIS-HEP Fellowship, 2021

Teaching Experience

- **Course Tutor for Graduate Computer Vision course CSCI-GA 2271 by Prof. Rob Fergus at NYU:** Organized materials for weekly sections, held office hours and advised student projects.
- **Course Assistant for Artificial Intelligence (CSCI-UA.472) & Data Management & Analysis (CSCI-UA.479) courses at NYU:** Assisted students with their homeworks & graded the programming & theory assignments.

Skills

Languages	Python, C++, C, CUDA, SQL, Latex
Frameworks	Pytorch, Tensorflow/Keras, Horovod, ONNX, TensorRt, MxNet
Development Tools	Git, Docker, Singularity
Libraries	Pandas, Numpy, OpenCV, Matplotlib, SciPy, Seaborn

Other Highlighted Projects

- **Making Every Model Robust:**
  - Built robust defense mechanism against white box adversarial attacks using natural supervision and quantization techniques agnostic to the victim ML model.
  - This method weakens the adversarial attacks on images generated by AutoAttack by restoring their natural structure and takes advantage of the quantized training of ML models.
  - The results of this method beat RobustBench benchmarks and were among the best achieved by everyone who had taken the graduate course of foundations of machine learning.